# Cognitive Debt: AI as Intellectual Leverage and the Dynamics of Systemic Fragility*

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*Preliminary — Comments Welcome*

**Abstract**

We develop a formal theory of *cognitive debt*: the stock of unverified reasoning obligations that accumulates when individuals use AI as a substitute—rather than a complement—for first-principles cognition. The model features two state variables per agent—*cognitive capital* (the unaided ability to reason, verify, and transfer knowledge) and *cognitive debt* (the corresponding unserviced obligation)—and a multiplicative production technology in which cognitive capital functions as collateral that determines the return to AI adoption. We establish five propositions. (1) Rational agents incur positive cognitive debt because the costs are deferred, partially external, and masked by short-run productivity

*We thank participants at ... for helpful comments. All errors are our own.

gains. (2) Tranquil periods lower subjective risk assessments, raise AI substitution intensity, and compound leverage—generating a *cognitive Minsky moment*: subjective risk falls while true systemic fragility rises. (3) Expected crisis losses are convex in the aggregate leverage ratio, so high-leverage economies are disproportionately fragile. (4) Post-crisis, output-target pressure produces a *false-correction loop*: agents rationally patch AI failures with more AI, ratcheting leverage upward and guaranteeing larger future crises. (5) The decentralised equilibrium over-adopts substitutive AI relative to the social optimum, owing to three unpriced externalities: systemic risk, cognitive public goods, and an arms-race externality. The optimal Pigouvian instrument is an AI-use tax indexed to the aggregate leverage ratio. (6) In a two-type heterogeneous-agent economy, high-cognitive-capital agents adopt AI more intensively, erode their capital faster, and—in a reversal of fortune—eventually hold *less* unaided cognitive capital than initially lower-skilled agents; heterogeneity also amplifies aggregate systemic risk via the convexity of crisis losses.

*Keywords:* cognitive debt; AI adoption; intellectual leverage; Minsky fragility; human capital; systemic risk; externalities.

*JEL Codes:* O33, E44, J24, D62, G01.

# 1 Introduction

**A productivity paradox.** Consider two findings from the same experimental programme. Noy and Zhang [2023] randomly assign mid-career professionals access to a large language model and find that treated workers complete writing tasks 37% faster, at quality rated higher by independent evaluators. Dell'Acqua et al. [2023], in a field experiment with Boston Consulting Group analysts, replicate the productivity gain for routine in-distribution tasks—and then ask workers to solve problems beyond the AI's training frontier. For these *out-of-distribution* tasks, the high-AI-use group performs *worse* than the control group: AI reliance has eroded the unaided judgment required to recognise when AI is unreliable.

The pattern is not an anomaly. It is the signature of a structural mechanism that this paper formalises: short-run productivity and long-run cognitive capacity are in tension, and agents who rationally maximise the former systematically underinvest in the latter. The result is a hidden stock of obligations—the difference between what agents can produce *with* AI and what they can produce *without* it. We call this stock *cognitive debt*.

**This paper.** We develop a dynamic general-equilibrium model of cognitive debt accumulation, systemic fragility, and welfare. The framework has two distinguishing features. First, cognitive capital functions as the *collateral* underlying AI adoption: the marginal product of AI is strictly proportional to unaided cognitive capacity, so capital erosion feeds back into rising AI dependence in a self-reinforcing cycle. Second, AI failure is a *common-mode* event: when AI systems share training data and users share workflows, errors are correlated, and model collapse [Shumailov et al., 2024] degrades AI quality in proportion to aggregate reliance. These two features together generate Minsky-style instability in the cognitive domain.

Our central contribution is to show that three individually benign features—which each appear either desirable or at worst neutral—combine to produce systemic fragility.

1. **Multiplicative complementarity.** Cognitive capital is not merely complementary to AI; it is the *base asset* whose level determines the return to AI use. High-capital agents extract more from AI; low-capital agents cannot effectively prompt, verify, or transfer AI outputs. This creates a feedback: capital erosion reduces the marginal product of AI, which does not reduce AI reliance—it increases it, because output targets remain fixed.

2. **Invisible debt.** Cognitive debt is unobservable in normal operation. Skill atrophy surfaces only when AI is unavailable, when tasks exceed the AI's capability frontier, or when AI produces confident hallucinations that require first-principles correction. In the model, this corresponds to the stress state: debt exposure $\kappa z b_{it}$ is visible only at shock intensity $z > 0$.

3. **Systemic correlation.** AI errors are not idiosyncratic. When agents adopt the same AI systems, their errors become correlated. When AI-generated content dominates training pipelines, model collapse [Shumailov et al., 2024] erodes AI quality over time. Both forces mean that AI failure is a common-mode event, not a diversifiable risk.

**Main results.** In a model with a continuum of agents choosing AI substitution intensity and deliberate-practice investment, we establish:

**Proposition 1** shows that rational agents optimally incur positive cognitive debt. The determinants of equilibrium debt are transparent and empirically observable: AI quality, output pressure, discount factors, the subjective probability of AI failure, and the individual's exposure to failure costs.

**Proposition 2** establishes the cognitive Minsky moment. Every tranquil period—a stretch without observed AI failures—updates subjective risk downward and raises equilibrium AI substitution intensity. Cognitive leverage $\Omega_t = \bar{B}_t/\bar{K}_t$ rises monotonically while the true crisis probability also rises. The Minsky divergence—$\hat{\pi}_t \downarrow$ while $\pi_t \uparrow$—is a necessary consequence of rational Bayesian updating in a system where the true risk process is endogenous to the belief-driven leverage cycle.

**Proposition 3** shows that crisis losses are convex in leverage. Small increases in $\Omega$ near the fragility threshold produce disproportionately large expected losses, because both the probability and the severity of a cognitive crisis are increasing in $\Omega$.

**Proposition 4** characterises the *false-correction loop*. When the shadow price of current output exceeds the shadow price of future cognitive capital recovery, the individually optimal response to an AI failure is to adopt *more* AI assistance—patching AI errors with AI. The condition is generically satisfied under competitive output pressure, generating ratchet dynamics in leverage.

**Proposition 5** derives the social planner's optimum and quantifies the three externalities that drive the decentralised gap: (i) a systemic risk externality, because my AI use raises the aggregate leverage ratio and the common-mode failure probability without my bearing the full loss; (ii) a cognitive public-goods externality, because societal verification capacity depends on the aggregate stock of unaided cognitive capital; and (iii) an arms-race externality, because rising average AI use shifts the competitive output benchmark upward, compelling others to increase AI reliance.

**Proposition 6** (heterogeneous agents) establishes a *reversal of fortune*: high-cognitive-capital agents adopt AI more intensively (since the return to AI is proportional to $k$), erode their capital faster, and eventually end up with *less* unaided cognitive capital than low-capital agents who adopted AI conservatively. In the long run, AI adoption compresses the distribution of cognitive capital even as it initially widens output inequality.

Figure 1 illustrates the Minsky divergence and leverage dynamics; Figure 2 shows the equilibrium trajectory through the Hedge–Speculative–Ponzi phase space; Figure 3 displays the heterogeneous-agent inequality reversal.

**Related literature.** Our paper lies at the intersection of four strands.

*Technology and human capital.* Acemoglu and Restrepo [2018] study automation as task displacement; Acemoglu and Restrepo [2019] introduce reinstatement tasks. We differ in focusing on the *intertemporal* erosion of the capital stock rather than the contemporaneous task allocation, and in introducing a debt state variable with its own compounding dynamics.

*Financial fragility.* The Minsky framework [Minsky, 1986] classifies financial positions by the relationship between income flows and debt obligations. Kiyotaki and Moore [1997] formalise collateral constraints in credit cycles. Bernanke et al. [1999] develop the financial accelerator. We import the Minsky classification into the cognitive domain, with cognitive capital playing the role of collateral and cognitive debt compounding through habit formation and rising switching costs.

*AI and productivity.* Brynjolfsson et al. [2025] and Noy and Zhang [2023] document short-run productivity gains from AI access. Crucially, Dell'Acqua et al. [2023] identify the *jagged frontier*: AI improves performance on in-distribution tasks but *reduces* it on out-of-distribution tasks for high-AI-reliance workers. This asymmetry is the empirical anchor for our stress-state production function.

*Systemic risk and externalities.* Allen and Gale [2000] and Acemoglu et al. [2015] study contagion in financial networks. Model collapse [Shumailov et al., 2024] is the cognitive analogue: AI-generated content in training pipelines degrades AI quality in a manner that is correlated across all users of a given model family.

**Organisation.** Section 2 presents the model. Section 3 solves the individual problem (Proposition 1). Section 4 characterises aggregate dynamics (Propositions 2 and 3). Section 5 analyses post-crisis dynamics (Proposition 4). Section 6 derives the social optimum and optimal policy (Proposition 5). Section 7 develops the two-type heterogeneous-agent economy (Proposition 6) and the reversal of fortune. Section 8 discusses extensions and limitations. Section 9 concludes. All proofs are in the Appendix.

# 2 The Model

## 2.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. There is a unit continuum of agents indexed by $i \in [0, 1]$. Each agent is characterised by two state variables:

- $k_{it} \geq 0$: *cognitive capital*—the unaided ability to reason, verify, synthesise, and transfer knowledge. This is the stock of intellectual competence that functions without AI support.

- $b_{it} \geq 0$: *cognitive debt*—the gap between current task demands and current cognitive capital, made viable by AI. Formally, $b_{it}$ is the stock of unverified reasoning obligations: outputs produced with AI assistance that the agent could not independently reproduce, verify, or correct.

Each period, agent $i$ chooses two controls:

- $a_{it} \in [0, 1]$: *AI substitution intensity*—the fraction of cognitively demanding steps outsourced to AI. We reserve the label *substitutive* AI for $a_{it} > 0$; AI used purely as a tutor, verifier, or feedback device that requires the agent to engage first-principles reasoning corresponds to $a_{it} = 0$.

- $x_{it} \geq 0$: *deliberate-practice investment*—cognitive debt repayment: hand-derivation, unaided testing, active verification, first-principles reconstruction.

## 2.2 Production Technology

Nature draws a state $s_t \in \{N, S\}$ each period, where $N$ is the normal state and $S$ is the stress state. The true probability of the stress state is $\pi_t$, which is determined endogenously in equilibrium (Section 4). Agent $i$ holds the subjective belief $\hat{\pi}_t$ about the probability of the stress state, which is updated via Bayes' rule as described below.

### Normal-state production

In state $N$, output is

$$y_{it}^N = k_{it} \cdot G(a_{it};\, q_t), \tag{1}$$

where $q_t > 0$ denotes the quality of the prevailing AI system.

**Assumption 1** (Normal-state technology)**.** $G : [0,1] \times \mathbb{R}_{++} \to [1, \infty)$ *is twice continuously differentiable and satisfies:*

(i) $G(0;\, q) = 1$ *for all* $q > 0$.

(ii) $G_a(a;\, q) > 0$ *and* $G_{aa}(a;\, q) < 0$ *for all* $(a, q)$.

(iii) $G_q(a;\, q) > 0$ *and* $G_{aq}(a;\, q) > 0$ *for all* $(a, q)$.

(iv) $\lim_{a \to 0} G_a(a;\, q) = +\infty$ (Inada).

Condition (i) normalises: without AI, output equals cognitive capital $k_{it}$. Condition (ii) says AI raises output at a diminishing marginal rate. Condition (iii) says higher AI quality raises both the level and the marginal product of AI use. Condition (iv) ensures an interior solution whenever the cost of AI use is finite.

**Remark 1** (Cognitive capital as collateral). *The multiplicative structure* $y^N = k_{it} \cdot G(a_{it}; q_t)$ *encodes the key mechanism of the paper. The marginal product of AI is* $\partial y^N/\partial a = k_{it} \cdot G_a(a_{it}; q_t)$*, which is strictly proportional to* $k_{it}$*. Thus cognitive capital is not merely an additive complement to AI; it is the* scale factor*—the base asset—that determines how much value an agent extracts from any given level of AI substitution. An agent with* $k = 0$ *obtains zero output regardless of* $a$*: she cannot prompt effectively, cannot evaluate responses, cannot transfer AI-generated results to novel tasks. This multiplicative complementarity generates the leverage dynamics central to the model.*

#### Stress-state production

In state $S$—an AI outage, a hallucination in a critical task, an out-of-distribution problem, an audit requiring first-principles justification, or any event that forces the agent to operate at or beyond the AI's capability frontier—output is

$$y_{it}^S = k_{it} \cdot \tilde{G}(a_{it}; q_t, z_t) - \kappa \, z_t \, b_{it}, \tag{2}$$

where $z_t > 0$ is the intensity of the stress shock drawn from a distribution with support $[\underline{z}, \bar{z}]$, and $\kappa > 0$ is the debt-exposure coefficient.

**Assumption 2** (Stress-state technology). $\tilde{G}(a; q, z) = G\big(a; q \cdot s(z)\big)$*, where* $s : \mathbb{R}_{++} \to (0, 1]$ *satisfies* $s(0^+) = 1$*,* $s'(z) < 0$*, and* $s(z) \to 0$ *as* $z \to \infty$*.*

Assumption 2 says the stress state reduces *effective AI quality* by a factor $s(z) \in (0, 1)$: the AI system is less reliable, its outputs require deeper verification, or the task has moved outside the training distribution. Under this specification, $\tilde{G}_a = G_a(\cdot; qs(z)) < G_a(\cdot; q)$, so AI assistance provides a strictly smaller marginal benefit in the stress state.

The term $\kappa z_t b_{it}$ is the *debt-service cost*: cognitive debt $b_{it}$ represents the accumulated gap between task demands and unaided cognitive capacity. Under stress, this gap is exposed. Agents who have outsourced their verification, derivation, and error-correction to AI cannot fulfil these obligations from their own cognitive resources; the resulting output shortfall is proportional to $b_{it}$ and scaled by shock severity $z_t$.

**Remark 2** (Jagged frontier)**.** *Dell'Acqua et al. [2023] document that AI improves performance for in-distribution tasks but reduces performance for out-of-distribution tasks, particularly for workers who rely heavily on AI. Equation* (2) *formalises this: at high $a_{it}$, the agent's unaided capacity $k_{it}$ is low (from learning atrophy), and the AI's effective quality $qs(z)$ is low. Both forces simultaneously reduce $y^S$.*

## 2.3 Cognitive Capital and Debt Dynamics

The state variables evolve according to:

$$k_{i,t+1} = (1-\delta)\,k_{it} + \ell(1-a_{it}) + \nu\,x_{it}, \tag{3}$$

$$b_{i,t+1} = (1+r_b)\,b_{it} + d(a_{it}) - \rho\,x_{it}, \tag{4}$$

where $\delta \in (0,1)$ is the cognitive depreciation rate, $r_b > 0$ is the cognitive debt compounding rate, $\nu > 0$ is the learning efficiency of deliberate practice, and $\rho > 0$ is its debt-reduction efficiency.

**Assumption 3** (Dynamics)**.** (i) *$\ell : [0,1] \to \mathbb{R}_+$ is $C^2$ with $\ell(0) = 0$, $\ell'(s) > 0$, $\ell''(s) \leq 0$. (Learning by doing is increasing and weakly concave in unaided effort fraction $1-a$.)*

(ii) *$d : [0,1] \to \mathbb{R}_+$ is $C^2$ with $d(0) = 0$, $d'(a) > 0$, $d''(a) > 0$. (Cognitive debt accumulation is increasing and strictly convex in AI substitution intensity.)*

(iii) $r_b > 0$. *(Cognitive debt compounds: foundational gaps raise the marginal cost of subsequent learning and error-detection.)*

The law of motion for $k$ reflects two learning channels. The term $\ell(1 - a_{it})$ is *learning by doing* from unaided cognition: working through a problem independently—even imperfectly—builds transferable capacity. When $a_{it} = 1$, this channel is shut: the agent never engages the reasoning process. The term $\nu x_{it}$ is *deliberate practice*: targeted effort to rebuild or maintain cognitive capacity.

The law of motion for $b$ reflects three debt dynamics. The term $(1 + r_b)b_{it}$ captures compounding: existing gaps make new gaps easier to acquire (habit formation, rising prompt dependency, declining verification skills). The term $d(a_{it})$ reflects new debt issuance, convex in $a_{it}$ because marginal substitution increasingly replaces higher-order reasoning. The term $-\rho x_{it}$ is debt repayment.

**Remark 3** (Hedge, Speculative, and Ponzi Cognition)**.** *Following Minsky [1986]'s financial classification, define:*

- Hedge cognition*: $k_{it} \geq b_{it} +$ task demand. The agent can service all cognitive obligations from own capital.*

- Speculative cognition*: $k_{it} <$ task demand, but $k_{it} +$ AI support $\geq$ task demand. The agent meets current demands only by rolling over AI reliance—paying "interest" but not principal.*

- Ponzi cognition*: $d(a_{it}) + r_b b_{it} > \rho x_{it}^{\max}$. New debt and compounding exceed maximum repayment capacity. The agent sustains output only by escalating $a_{it}$, constituting a cognitive Ponzi scheme.*

*The central theoretical result (Proposition 2) is that the decentralised equilibrium systematically migrates agents from hedge through speculative to Ponzi cognition during tranquil periods.*

## 2.4 AI Quality and Model Collapse

AI quality $q_t$ is determined at the aggregate level. When agents extensively use AI-generated content, two degradation forces operate.

**Assumption 4** (AI quality dynamics)**.**

$$q_{t+1} = \bar{q} \cdot (1 - \gamma \bar{A}_t) \cdot (1 + \eta\, \Omega_t)^{-1}, \tag{5}$$

*where* $\bar{A}_t = \int_0^1 a_{it}\, \mathrm{d}i$*,* $\Omega_t = \bar{B}_t / \bar{K}_t$ *is the aggregate cognitive leverage ratio,* $\gamma > 0$*, and* $\eta > 0$*.*

The first factor $(1 - \gamma \bar{A}_t)$ captures *model collapse* [Shumailov et al., 2024]: when AI-generated content enters training pipelines at scale, model quality degrades because the diversity and grounding of the training distribution shrinks. The second factor $(1 + \eta \Omega_t)^{-1}$ captures *verification failure*: as aggregate leverage $\Omega_t$ rises, fewer agents retain the unaided capacity to detect and correct AI errors, so bugs, hallucinations, and systematic biases propagate unfiltered into knowledge stocks.

## 2.5 Information Structure and Belief Dynamics

Agents are rational Bayesians. They observe their own output and whether a crisis occurs, but they do not observe the aggregate leverage ratio $\Omega_t$ or the true crisis probability $\pi_t$. The subjective probability of the stress state is updated by:

$$\hat{\pi}_{t+1} = (1 - \lambda)\, \hat{\pi}_t + \lambda \cdot \mathbf{1}\{\text{crisis at } t\}, \tag{6}$$

where $\lambda \in (0, 1)$ is the learning rate. In a period without a crisis, $\hat{\pi}_{t+1} < \hat{\pi}_t$: subjective risk falls. This is the empirically natural specification—agents update on realised events—and it generates the Minsky divergence characterised in Proposition 2.

The true crisis probability is:

$$\pi_t = \Pi(\Omega_t), \qquad \Pi' > 0,\ \Pi'' \geq 0,\ \Pi(0) = 0,\ \lim_{\Omega \to \infty} \Pi(\Omega) = 1. \tag{7}$$

We adopt the parametric form $\Pi(\Omega) = 1 - e^{-\lambda_\pi \Omega^\gamma}$ with $\gamma \geq 1$ for the proofs that require explicit computation; the qualitative results hold for any $\Pi$ satisfying (7).

# 3 Individual Optimization

## 3.1 The Agent's Problem

Agent $i$ maximises discounted expected utility, taking as given the AI quality path $\{q_t\}$, aggregate variables $\{\Omega_t\}$, and the subjective crisis probability $\hat{\pi}_t$:

$$V(k, b;\, \hat{\pi}, q) = \max_{a \in [0,1],\, x \geq 0} \left\{ (1 - \hat{\pi})\, y^N(k, a; q) + \hat{\pi}\, \mathbb{E}_z \big[ y^S(k, a, b;\, q, z) \big] - c(x) + \beta\, \mathbb{E} \big[ V(k', b';\, \hat{\pi}', q') \big] \right\}, \tag{8}$$

where $c(x)$ is the direct cost of deliberate practice with $c(0) = 0$, $c'(x) > 0$, $c''(x) > 0$; $\beta \in (0, 1)$ is the discount factor; and $k', b'$ follow (3)–(4).

To characterise individual incentives in closed form and derive comparative statics, we work with a two-period version of the problem. The infinite-horizon problem yields qualitatively identical results (Appendix G).

### Two-period analytical model

An agent lives for two periods, $t = 0$ and $t = 1$. In $t = 0$ she chooses $a$ and $x$. In $t = 1$ she consumes the residual value of her state. The period-1 value function is:

$$V_1(k', b') = \mu_k\, k' - \mu_b\, b', \tag{9}$$

where $\mu_k > 0$ and $\mu_b > 0$ are shadow values of cognitive capital and debt in the terminal period (derived from the infinite-horizon envelope conditions in Appendix I). The objective is:

$$\max_{a,x} \underbrace{(1-\hat{\pi})\, kG(a;q) + \hat{\pi}\, \mathbb{E}_z\big[k\tilde{G}(a;q,z) - \kappa z b\big]}_{\text{expected current output}} - c(x) + \beta\, V_1(k', b'). \tag{10}$$

Expanding using (3)–(4) and (9):

$$\max_{a,x} \big[(1-\hat{\pi})\, kG(a;q) + \hat{\pi}\, \mathbb{E}_z[k\tilde{G}(a;q,z)]\big] - \hat{\pi}\, \kappa\, \bar{z}\, b - c(x) + \beta\, \mu_k\, \big[(1-\delta)k + \ell(1-a) + \nu x\big] - \beta\, \mu_b\, \big[(1+r_b)b + d(a) - \tag{11}$$

where $\bar{z} = \mathbb{E}[z \mid S]$.

**First-order conditions**

Differentiating (11) with respect to $a$ yields the first-order condition for AI substitution:

$$\boxed{k\, \big[(1-\hat{\pi})\, G_a(a^*;q) + \hat{\pi}\, \tilde{G}_a(a^*;q,\bar{z})\big] = \beta\, \big[\mu_k\, \ell'(1-a^*) + \mu_b\, d'(a^*)\big].} \tag{12}$$

The left-hand side is the expected marginal product of AI in current output. The right-hand side is the discounted marginal future cost: forgone learning by doing (valued at $\mu_k$) plus newly issued cognitive debt (valued at $\mu_b$).

Differentiating with respect to $x$ yields the condition for deliberate practice:

$$c'(x^*) = \beta\, (\mu_k\, \nu + \mu_b\, \rho). \tag{13}$$

Practice is chosen until its marginal cost equals the discounted return: increasing cognitive capital (at rate $\nu$, valued at $\mu_k$) plus reducing cognitive debt (at rate $\rho$, valued at $\mu_b$).

## 3.2 Proposition 1: Rational Cognitive Debt

**Proposition 1** (Rational Cognitive Debt)**.** *Under Assumptions 1–3, for any* $\hat{\pi} \in [0, 1)$*,* $q > 0$*, and* $k > 0$*:*

(i) *There exists a unique interior solution* $a^* \in (0, 1)$ *to* (12)*, and the resulting cognitive debt issuance satisfies* $d(a^*) > 0$*.*

(ii) *The equilibrium AI substitution intensity satisfies:*

$$\frac{\partial a^*}{\partial q} > 0, \quad \frac{\partial a^*}{\partial k} > 0, \quad \frac{\partial a^*}{\partial \hat{\pi}} < 0, \quad \frac{\partial a^*}{\partial \beta} < 0, \quad \frac{\partial a^*}{\partial \mu_b} < 0, \quad \frac{\partial a^*}{\partial \kappa} < 0.$$

(iii) *The "cognitive debt wedge" satisfies:*

$$a^*(\hat{\pi}) < a^{\textit{no-debt}}$$

*where* $a^{\textit{no-debt}}$ *is the optimal AI use when* $d \equiv 0$*, confirming that debt accumulation is the mechanism restraining (but not eliminating) AI substitution.*

*Proof.* See Appendix B. □

**Remark 4** (Economic interpretation of comparative statics)**.** *Part (ii) translates directly into policy-relevant predictions:*

- $\partial a^*/\partial q > 0$*: AI capability improvements raise equilibrium debt issuance—a form of "keeping up" with the technology that progressively erodes the cognitive base.*
- $\partial a^*/\partial k > 0$*: Agents with higher cognitive capital use AI more intensively (the multiplicative return is higher), generating higher gross debt. This creates a non-monotone inequality trajectory: high-*$k$ *agents gain more short-run productivity but face larger potential capital erosion.*

- $\partial a^*/\partial\hat{\pi} < 0$*: Higher perceived stress probability deters AI substitution. This is the "insurance" channel: agents who believe AI may fail invest in maintaining their unaided capacity.*

- $\partial a^*/\partial\beta < 0$*: More patient agents internalise more of the future cost of cognitive capital erosion. Short-termism is a direct driver of excessive AI reliance.*

# 4 Aggregate Dynamics and the Cognitive Minsky Moment

## 4.1 Competitive Equilibrium

**Definition 1** (Competitive Equilibrium)**.** *A competitive equilibrium is a sequence of individual policies* $\{a_{it}^*, x_{it}^*\}$*, aggregate variables* $\{K_t, B_t, A_t, \Omega_t\}$*, AI quality* $\{q_t\}$*, and belief sequences* $\{\hat{\pi}_t, \pi_t\}$ *such that:*

(i) *Each agent solves* (8) *taking* $\{q_t, \hat{\pi}_t\}$ *as given.*

(ii) *Markets clear:* $K_t = \int k_{it}\,\mathrm{d}i$, $B_t = \int b_{it}\,\mathrm{d}i$, $A_t = \int a_{it}\,\mathrm{d}i$, $\Omega_t = B_t/K_t$.

(iii) *AI quality follows* (5).

(iv) *Beliefs follow* (6) *and* (7).

We focus on symmetric equilibria in which $a_{it}^* = a_t^*$ and $x_{it}^* = x_t^*$ for all $i$, so $K_t = k_t$ and $B_t = b_t$ in the representative-agent characterisation. Heterogeneous-agent extensions are discussed in Section 8.

## 4.2 Proposition 2: The Cognitive Minsky Moment

**Proposition 2** (Cognitive Minsky Moment)**.** *Fix AI quality $q$ and suppose a tranquil period $\mathcal{T} = \{0, 1, \ldots, T-1\}$ in which no crisis is realised. Then along any equilibrium path through $\mathcal{T}$:*

(i) *Subjective risk is strictly decreasing: $\hat{\pi}_{t+1} < \hat{\pi}_t$ for all $t \in \mathcal{T}$.*

(ii) *Equilibrium AI substitution intensity is strictly increasing: $a^*_{t+1} > a^*_t$ for all $t \in \mathcal{T}$.*

(iii) *Aggregate cognitive leverage is strictly increasing: $\Omega_{t+1} > \Omega_t$ for all $t \in \mathcal{T}$.*

(iv) *True crisis probability is strictly increasing: $\pi_{t+1} = \Pi(\Omega_{t+1}) > \Pi(\Omega_t) = \pi_t$ for all $t \in \mathcal{T}$.*

(v) *There exists $T^* < \infty$ such that for all $t > T^*$, $\hat{\pi}_t < \pi_t$* (the Minsky divergence)*: perceived risk falls below true systemic risk.*

*Proof.* See Appendix C. $\square$

**Remark 5** (Minsky divergence and rational expectations)**.** *The divergence $\hat{\pi}_t < \pi_t$ (part (v)) is consistent with rational Bayesian updating because $\pi_t$ is determined by the* endogenous *leverage process, which agents do not observe directly. Agents update correctly on realised crises but cannot infer the true $\pi_t$ from the absence of crises, because each crisis-free period is also evidence that the current leverage level has not yet triggered an event. The divergence is therefore not a sign of irrationality; it is a consequence of the unobservability of the aggregate state.*

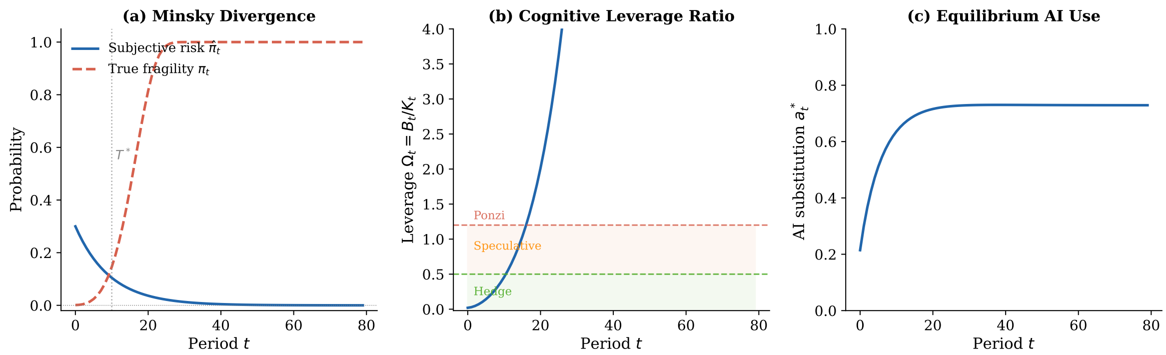


Figure 1: **Minsky Divergence and Cognitive Leverage Dynamics.** Simulated equilibrium path under the parametric model ($q = 1$, $\phi = 0.35$, $\eta = 0.60$, $\delta = 0.04$, $r_b = 0.10$, $\beta = 0.95$, $\hat{\pi}_0 = 0.30$, no crises realised for $T = 80$ periods). *Panel (a):* Subjective risk $\hat{\pi}_t$ falls monotonically while true fragility $\pi_t$ rises; the vertical dotted line marks the crossing point $T^*$. *Panel (b):* The aggregate cognitive leverage ratio $\Omega_t = B_t/K_t$ drifts upward through the Hedge, Speculative, and Ponzi zones (defined at $\Omega = 0.50$ and $\Omega = 1.20$ respectively). *Panel (c):* Equilibrium AI substitution intensity $a_t^*$ rises as subjective risk falls, closing the feedback loop.

## 4.3 Proposition 3: Convexity of Crisis Losses

Define the expected crisis loss at date $t$ as:

$$L_t = \Pi(\Omega_t) \cdot \kappa \, \mathbb{E}_z\big[\max\{0,\, z\, B_t - \mathcal{V}(K_t)\}\big], \tag{14}$$

where $\mathcal{V}(K) = K^\alpha$ ($\alpha \in (0,1)$) is the aggregate verification capacity—the concave function representing society's ability to detect and correct AI errors as a function of aggregate cognitive capital.

**Proposition 3** (Convex Fragility)**.** *Under Assumptions 1–4, the expected crisis loss $L_t$ is:*

(i) *Strictly increasing in the aggregate leverage ratio: $\partial L_t/\partial\Omega_t > 0$.*

(ii) *Strictly convex in the aggregate leverage ratio: $\partial^2 L_t/\partial\Omega_t^2 > 0$ in a neighbourhood of any interior $\Omega_t$.*

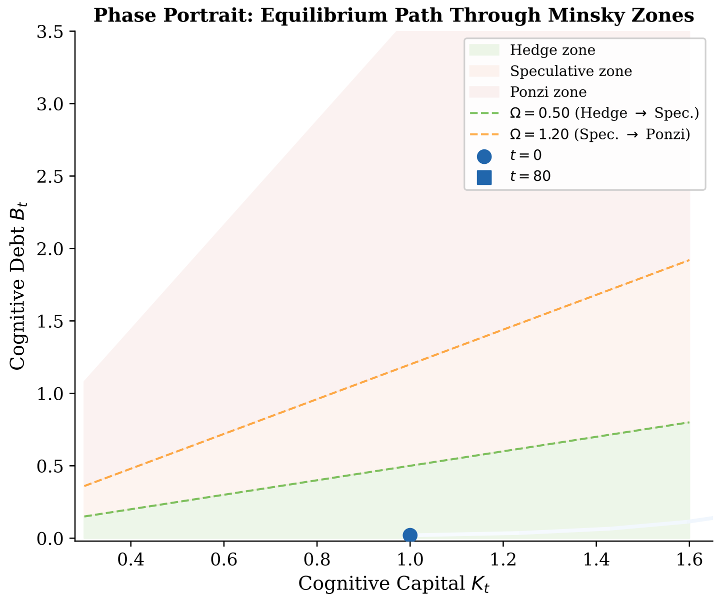


Figure 2: **Phase Portrait in Cognitive Capital–Debt Space.** The equilibrium trajectory (blue gradient line, darker = later) starts in the Hedge zone ($\Omega < 0.50$) and migrates through the Speculative zone into the Ponzi zone during an uninterrupted tranquil period. The colour intensity encodes time: early periods are light blue, late periods are dark blue. Arrows indicate the direction of motion. Iso-leverage lines at $\Omega = 0.50$ (green dashed) and $\Omega = 1.20$ (orange dashed) delimit the three Minsky zones.

(iii) *Strictly increasing in AI model concentration $M_t$ (the HHI of AI system market share): $\partial L_t / \partial M_t > 0$.*

*Proof.* See Appendix D. □

**Remark 6** (Why convexity matters)**.** *Convexity of $L_t$ in $\Omega_t$ implies that small additional increments of cognitive leverage, once leverage is already high, produce disproportionately large expected losses. This gives a formal rationale for the precautionary principle: a society near the fragility threshold should weight leverage reduction much more than one near the safe zone, even if their current expected losses are similar.*

# 5 Post-Crisis Dynamics: The False-Correction Loop

## 5.1 Setup

Suppose a stress event occurs at date $t = 0$: the stress state is realised with intensity $z_0 > 0$. Agents observe the shortfall in their unaided capacity. Do they reduce AI substitution intensity—repaying cognitive debt—or do they increase it to maintain output?

Let $w > 0$ denote the competitive output target (the market-determined minimum output level required for continued participation in the labour market). Agents face a binding output constraint:

$$y_{i0} \geq w. \tag{15}$$

This constraint captures the competitive pressure documented in the literature: AI adoption by some agents shifts the performance distribution, raising the benchmark against which all agents are evaluated [Brynjolfsson et al., 2025].

**Definition 2** (False-Correction Loop)**.** *A* false-correction loop *occurs when, following a crisis at $t = 0$, the equilibrium AI substitution intensity satisfies $a_1^* > a_0^-$, where $a_0^-$ denotes the pre-crisis substitution intensity, and when the resulting leverage $\Omega_1 > \Omega_0^-$.*

**Proposition 4** (False-Correction Loop)**.** *Suppose the output constraint* (15) *is binding at $t = 1$ and that $k_1 < k_0^-$ (cognitive capital has declined through the crisis period). Then the post-crisis equilibrium AI substitution intensity satisfies $a_1^* > a_0^-$ if and only if:*

$$\underbrace{\lambda_y \, k_1 \, G_a(a_0^-; q)}_{\text{shadow value of current output}} \quad > \quad \underbrace{\beta \left[\mu_k \, \ell'(1 - a_0^-) + \mu_b \, d'(a_0^-)\right]}_{\text{shadow value of cognitive capital recovery}} \; . \tag{16}$$

*When condition* (16) *holds, the post-crisis path satisfies $\Omega_{t+1} > \Omega_t$ for all $t \geq 1$ until a new crisis occurs, with successive crises of non-decreasing severity.*

*Proof.* See Appendix E. □

**Remark 7** (When does the false-correction loop operate?). *Condition* (16) *is more likely to be satisfied when: (a) the output target $w$ is high (high $\lambda_y$); (b) cognitive capital $k_1$ is low following the crisis (paradoxically making the condition harder to satisfy in the left-hand side, but also reducing the option value of capital recovery); (c) the discount factor $\beta$ is low (short-termism); (d) the debt compounding rate $r_b$ is high (making capital recovery expensive). In practice, the condition is most likely binding in competitive labour markets where output is continuously monitored and the opportunity cost of a period of reduced output is high.*

**Remark 8** (Financial analogy). *The false-correction loop is the cognitive analogue of the "leveraged buyout of a failing firm": when the firm cannot service its existing debt, the rational response in a competitive product market is to borrow more to maintain operations, rolling over obligations until eventual insolvency. Here, the firm is the agent's cognitive system, the debt is cognitive debt, and the operations are current output production. The loop is individually rational but socially destructive.*

# 6 Welfare Analysis

## 6.1 Three Externalities

The decentralised equilibrium exhibits three distinct externalities:

**1. Systemic risk externality.** When agent $i$ raises $a_{it}$, the aggregate leverage $\Omega_t$ rises. This increases the true crisis probability $\pi_t = \Pi(\Omega_t)$ and the expected aggregate loss $L_t$. Agent $i$ does not bear the full marginal social cost:

$$\frac{\partial L_t}{\partial a_{it}} \cdot \frac{1}{\text{mass of agents}} < \frac{\partial L_t}{\partial \Omega_t} \cdot \frac{\partial \Omega_t}{\partial a_{it}}. \tag{17}$$

**2. Cognitive public-goods externality.** Society's verification capacity $\mathcal{V}(K_t)$ is a public good: all agents benefit from a high-$K_t$ population that can scrutinise AI outputs, train the next generation, and provide error-correction in shared knowledge domains. When agent $i$ substitutes AI for cognition, $K_t$ falls, degrading this public good. The private return to $a_{it}$ does not account for this spillover.

**3. Arms-race externality.** Competitive output benchmarking means that when agent $i$ raises output via AI, the competitive standard rises for all other agents, forcing them to raise their own $a_{jt}$. This is a coordination failure: all agents would prefer the equilibrium with lower $a$, but individually each has an incentive to deviate upward.

## 6.2 The Social Planner's Problem

A benevolent social planner chooses aggregate $\{A_t, X_t\}$ to maximise the sum of agents' welfare, internalising all externalities:

$$\max_{\{A_t, X_t\}} \sum_{t=0}^{\infty} \beta^t \left\{ (1-\pi_t)\, K_t\, G(A_t;\, q_t) + \pi_t\, \mathbb{E}_z\big[K_t\, \tilde{G}(A_t;\, q_t, z) - \kappa z B_t\big] - C(X_t) \right\}, \quad (18)$$

subject to (3)–(4) in aggregate, (5), and (7), taking $\pi_t = \Pi(\Omega_t)$ as endogenous.

**Proposition 5** (Decentralised Over-adoption)**.** *Under Assumptions 1–4, the decentralised equilibrium aggregate AI substitution intensity exceeds the social optimum:*

$$A_t^D > A_t^P \quad \textit{for all } t \geq 0.$$

*The gap* $\Delta_t \equiv A_t^D - A_t^P$ *is strictly increasing in: (i) aggregate model concentration* $M_t$*; (ii) the length of the current tranquil period; and (iii) the convexity parameter of the debt accumulation function* $d$*.*

*The constrained-optimal Pigouvian tax on AI substitution intensity is:*

$$\tau_t^* = \underbrace{\beta\,\Pi'(\Omega_t)\,\frac{\partial\Omega_t}{\partial A_t}\,L_t/\Pi(\Omega_t)}_{\text{systemic risk externality}} + \underbrace{\beta\,\mu_k\,\frac{\partial K_{t+1}}{\partial A_t}}_{\text{public-goods externality}} + \underbrace{\lambda_y\,\frac{\partial\bar{y}_t}{\partial a_{it}}}_{\text{arms-race externality}}, \tag{19}$$

*evaluated at the social optimum* $(A_t^P, X_t^P)$*. The optimal tax is (a) increasing in aggregate leverage* $\Omega_t$*, (b) higher when model concentration is higher, and (c) counter-cyclical—rising during tranquil periods as leverage accumulates.*

*Proof.* See Appendix F. □

**Remark 9** (Policy instruments)**.** *Equation* (19) *suggests several implementable policies. A levy indexed to the aggregate AI usage share (approximating* $\Omega_t$*) serves as the systemic risk tax. Mandatory unaided assessment requirements—professional certification exams taken without AI, stress-test protocols for AI-dependent workflows—serve as forced debt repayment (*$x_t$*). Limits on AI model market concentration address* $M_t$*. The counter-cyclicality result provides a formal rationale for tightening these requirements during periods of rapid AI adoption, analogous to counter-cyclical capital buffers in macroprudential regulation.*

# 7 Heterogeneous Agents and the Reversal of Fortune

## 7.1 Setup

We enrich the model with two types of agents: type $H$ (high initial cognitive capital, $k_{H,0} > \bar{k}$) and type $L$ (low initial cognitive capital, $k_{L,0} < \bar{k}$), in equal proportions. Both types begin with zero cognitive debt ($b_{H,0} = b_{L,0} = 0$) and the same subjective risk belief $\hat{\pi}_0$. All structural parameters are identical across types.

From Proposition 1, part (ii), $\partial a^*/\partial k > 0$: higher cognitive capital raises the return to AI (via the multiplicative structure $\partial y^N/\partial a = k \cdot G_a$), so the high-$k$ type adopts AI more intensively in equilibrium:

$$a^*_{H,t} > a^*_{L,t} \quad \text{for all } t \text{ such that } k_{H,t} > k_{L,t}. \tag{20}$$

## 7.2 Proposition 6: Reversal of Fortune

Define the *cognitive capital gap* $\Delta k_t \equiv k_{H,t} - k_{L,t}$ and the *AI adoption gap* $\Delta a^*_t \equiv a^*_{H,t} - a^*_{L,t} > 0$.

**Proposition 6** (Reversal of Fortune)**.** *In the two-type economy, during a tranquil period with no crises:*

(i) $\Delta k_t$ *is strictly decreasing for all* $t \geq 0$*: high-*$k$ *agents erode their cognitive capital advantage faster than low-*$k$ *agents.*

(ii) $\Delta a^*_t$ *is initially positive but converges to zero as* $\Delta k_t \to 0$*.*

(iii) *If the learning-by-doing differential satisfies*

$$\ell'(\xi)\, \Delta a^*_t > (1-\delta)\, \Delta k_t \quad \textit{for all } t \geq T^\dagger, \tag{21}$$

*for some* $T^\dagger < \infty$*, then there exists* $T^{**} < \infty$ *such that* $\Delta k_{T^{**}} = 0$ *and* $\Delta k_t < 0$ *for all* $t > T^{**}$*:* the high-$k$ type ends up with less cognitive capital than the low-$k$ type.

(iv) *The aggregate crisis loss in the heterogeneous economy exceeds that in the homo-*

*geneous economy with $\bar{k} = (k_{H,0} + k_{L,0})/2$:*

$$L_t^{hetero} > L_t^{homo},$$

*because the high-$k$ type's elevated leverage generates a disproportionate increase in expected losses (by convexity of $L$ in $\Omega$, Proposition 3).*

*Proof.* See Appendix H. □

**Remark 10** (Interpretation)**.** *Part (iii) is the central result. It says that the agents who benefit* most *from AI adoption in the short run—because their high $k$ makes the multiplicative return to $a$ large—accumulate cognitive debt fastest, erode their capital most severely, and may eventually fall below the initial laggards. The mechanism is entirely within the rational-agent framework: each type is optimising correctly given private information. The reversal is a general-equilibrium consequence of the multiplicative structure and the compounding debt dynamics.*

*Part (iv) implies that inequality is not merely a distributional concern: it is a source of additional systemic fragility. A more dispersed distribution of cognitive capital (for given mean) produces higher aggregate crisis losses, providing a novel rationale for policies that maintain a minimum floor of unaided cognitive capacity.*

# 8 Discussion

## 8.1 Heterogeneity and Inequality

Proposition 6 establishes that the reversal of fortune is not a knife-edge result: it holds whenever condition (21) is satisfied, which is generically true when the learning-by-doing differential is sufficiently large relative to the remaining capital gap. The

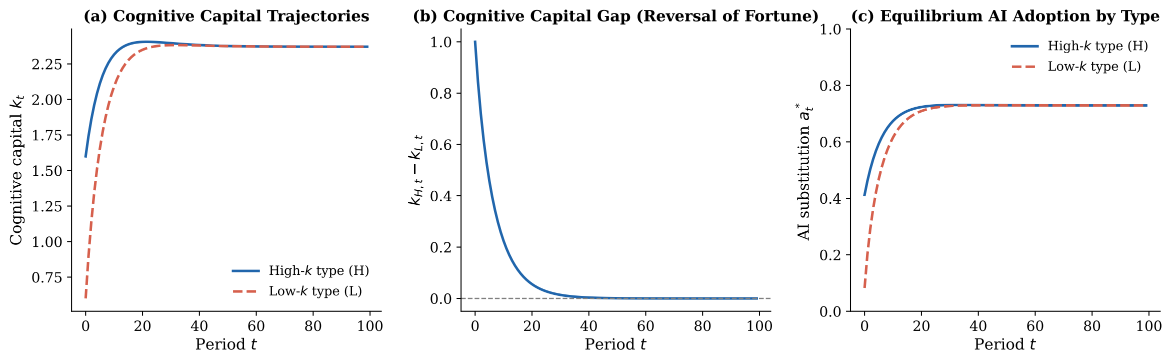


Figure 3: **Cognitive Capital Inequality and the Reversal of Fortune.** Simulated two-type economy ($k_{H,0} = 1.6$, $k_{L,0} = 0.6$; all other parameters as in Figure 1). *Panel (a):* Cognitive capital trajectories: the high-$k$ type (blue solid) declines faster and eventually converges toward—and crosses—the low-$k$ type (red dashed). *Panel (b):* The cognitive capital gap $k_{H,t} - k_{L,t}$ falls from its initial positive value, crosses zero at $T^{**}$ (vertical dotted line), and becomes negative—the reversal of fortune. *Panel (c):* AI adoption intensity: the high-$k$ type always adopts AI more intensively (Proposition 1, $\partial a^*/\partial k > 0$), which is the direct driver of the faster capital erosion.

parametric simulations in Figure 3 illustrate a concrete instance. The result has two policy implications. First, policies that protect the cognitive capital floor (mandatory unaided-practice requirements, AI-free examinations) disproportionately benefit high-adopting, high-skill agents who face the strongest erosion pressure. Second, the distributional reversal implies that standard skill-biased-technological-change frameworks—which predict widening inequality—may mischaracterise the long-run distribution of cognitive capital under AI adoption.

## 8.2 Complementary vs. Substitutive AI

The model throughout treats $a_{it}$ as substitutive AI use. Complementary AI use—Socratic tutoring, friction-generating feedback, forced verification—appears in the model as negative $d(a)$ (debt repayment) and positive $\ell$ (learning activation). The model therefore nests complementary AI use as a special case with reversed dynamics: complementary AI raises $k$ and reduces $b$. The key policy question is how to distinguish substitutive from complementary AI use in practice; the model provides a formal criterion: an AI interaction is complementary if it increases the agent's ability to perform the relevant task unaided.

## 8.3 The "Cognitive Reserve" Policy

An analogy from financial regulation is the *capital adequacy ratio*: banks must hold a minimum ratio of capital to risk-weighted assets. The cognitive analogue is a *cognitive reserve requirement*: a minimum ratio $k_{it}/b_{it} \geq \underline{\Omega}^{-1}$. When this constraint binds, agents must either reduce debt (through deliberate practice) or reduce issuance (by reducing $a_{it}$). This constraint would prevent Ponzi cognition as defined in Remark 3. Formal analysis of the transition dynamics under such a constraint is left for future work.

## 8.4 Limitations

Several simplifications warrant acknowledgement. First, we treat the cognitive capital state variable as one-dimensional; in reality, cognitive skills are domain-specific and heterogeneous. Second, the model assumes that the production function $G$ is known and stationary; in a world where AI capabilities are rapidly evolving, $G$ shifts over time and the appropriate value of $a$ changes accordingly. Third, we abstract from the possibility that AI may be used to enhance cognitive capital formation—the "Uzawa-learning" channel [Uzawa, 1965, Jr., 1988]. Incorporating endogenous AI quality into the human capital accumulation equation is a promising extension. Fourth, the model does not account for the possibility that some forms of cognitive capital—particularly tacit, embodied, and physical knowledge [Aschenbrenner, 2024]—may be non-reclaimable once lost.

# 9 Conclusion

We have developed a formal theory of cognitive debt: the unobserved stock of reasoning obligations that accumulates when individuals systematically outsource first-principles cognition to AI. The model shows that this debt arises from rational optimisation, compounds through habit formation and competitive pressure, and generates Minsky-style fragility: tranquil periods of AI-driven productivity systematically build the conditions for cognitive crises.

The five propositions identify the key structural forces. Rational agents borrow cognitive debt because the costs are deferred and partially externalised. Aggregate leverage rises during tranquil periods because belief updating and output competition both push toward more AI use. Crisis losses are convex in leverage, so the tail risk is large. Post-crisis adjustment follows the false-correction loop rather than genuine deleveraging. And the decentralised equilibrium over-adopts AI relative to the social optimum by a gap that grows during boom periods and is larger in more concentrated AI markets.

The policy implications are concrete: an AI-use tax indexed to aggregate leverage; mandatory unaided-performance audits; limits on AI model concentration; and deliberate-practice requirements for professionals in high-stakes domains. These instruments map directly to the three externalities identified in the welfare analysis.

More broadly, the framework offers a formal vocabulary for a concern that has been articulated informally but not yet analytically: that the short-run productivity gains from AI adoption may be purchased against a long-run degradation of the cognitive capacity required to verify, correct, extend, and ultimately reproduce those gains. Whether this trade-off is empirically large is an open question. The theory provides the structure within which to answer it.

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# Appendix: Proofs

## B Proof of Proposition 1

We prove each part in turn.

**Part (i): Existence and uniqueness of interior solution.** Define the FOC residual:

$$H(a;\,\hat{\pi},q,k,\mu_k,\mu_b) \equiv k\left[(1-\hat{\pi})\,G_a(a;q)+\hat{\pi}\,\tilde{G}_a(a;q,\bar{z})\right]-\beta\left[\mu_k\,\ell'(1-a)+\mu_b\,d'(a)\right]=0. \tag{22}$$

*At $a=0$:* By Assumption 1(iv), $G_a(0;q)=+\infty$. Since $\tilde{G}_a(0;q,\bar{z})=G_a(0;qs(\bar{z}))\cdot s(\bar{z})=+\infty$ (for any $s>0$), the left-hand side of (22) is $+\infty$. The right-hand side is finite by Assumption 3(i)(ii). Hence $H(0;\cdot)>0$.

*At $a=1$:* $G_a(1;q)$ and $\tilde{G}_a(1;q,\bar{z})$ are finite by Assumption 1. The right-hand side involves $\ell'(0)\geq 0$ and $d'(1)>0$, so the right-hand side is positive. Given the concavity of $G$ and $\ell$, $H(1;\cdot)$ may be negative (i.e., costs exceed benefits at $a=1$). Specifically, as $d''(a)>0$ and $G_{aa}<0$, $H$ is strictly decreasing in $a$ for $a$ sufficiently large. By the intermediate value theorem, there exists $a^*\in(0,1)$ with $H(a^*;\cdot)=0$.

*Uniqueness:* The second-order condition requires $\partial H/\partial a<0$.

$$\frac{\partial H}{\partial a}=k\left[(1-\hat{\pi})\,G_{aa}+\hat{\pi}\,\tilde{G}_{aa}\right]+\beta\,\mu_k\,\ell''(1-a)-\beta\,\mu_b\,d''(a). \tag{23}$$

The first term is strictly negative (Assumption 1(ii)). The second term is non-positive (Assumption 3(i)). The third term is strictly negative (Assumption 3(ii)). Hence $\partial H/\partial a<0$ globally, confirming strict concavity of the objective and uniqueness of $a^*$.

Since $d(0)=0$ and $d'(a)>0$ with $a^*>0$, we have $d(a^*)>0$. □

**Part (ii): Comparative statics.** By the implicit function theorem, for any parameter $\theta$:

$$\frac{\partial a^*}{\partial \theta} = -\frac{\partial H/\partial \theta}{\partial H/\partial a}.$$

Since $\partial H/\partial a < 0$ (from part (i)), the sign of $\partial a^*/\partial \theta$ equals the sign of $\partial H/\partial \theta$.

*$\partial a^*/\partial q > 0$:*

$$\frac{\partial H}{\partial q} = k\left[(1-\hat{\pi})\, G_{aq}(a^*; q) + \hat{\pi}\, \tilde{G}_{aq}(a^*; q, \bar{z})\right] > 0, \tag{24}$$

since $G_{aq} > 0$ by Assumption 1(iii) and $\tilde{G}_{aq} = G_{aq}(\cdot;\, qs)s > 0$. □

*$\partial a^*/\partial k > 0$:*

$$\frac{\partial H}{\partial k} = (1-\hat{\pi})\, G_a(a^*; q) + \hat{\pi}\, \tilde{G}_a(a^*; q, \bar{z}) > 0. \tag{25}$$

Both terms are strictly positive. □

*$\partial a^*/\partial \hat{\pi} < 0$:*

$$\frac{\partial H}{\partial \hat{\pi}} = k\left[\tilde{G}_a(a^*; q, \bar{z}) - G_a(a^*; q)\right] - \beta\, \kappa\, \bar{z}\, d'(a^*). \tag{26}$$

Under Assumption 2, $\tilde{G}_a < G_a$, so the first term is strictly negative. The second term is strictly negative. Hence $\partial H/\partial \hat{\pi} < 0$ and $\partial a^*/\partial \hat{\pi} < 0$. □

*$\partial a^*/\partial \beta < 0$:*

$$\frac{\partial H}{\partial \beta} = -\left[\mu_k\, \ell'(1-a^*) + \mu_b\, d'(a^*)\right] < 0.\square$$

*$\partial a^*/\partial \mu_b < 0$:*

$$\frac{\partial H}{\partial \mu_b} = -\beta\, d'(a^*) < 0.\square$$

*$\partial a^*/\partial \kappa < 0$:* The parameter $\kappa$ enters through the stress-state expected cost. Increasing $\kappa$ increases the marginal cost of debt, raising $\mu_b$ (by the envelope theorem on

the continuation value). Since $\partial a^*/\partial \mu_b < 0$, $\partial a^*/\partial \kappa < 0$. $\square$

**Part (iii): Cognitive debt wedge.** When $d \equiv 0$ (no debt accumulation), the FOC reduces to:

$$k\,[(1-\hat{\pi})\,G_a + \hat{\pi}\,\tilde{G}_a] = \beta\,\mu_k\,\ell'(1-a^{\text{no-debt}}).$$

When $d > 0$, the right-hand side of (12) gains the term $\beta\,\mu_b\,d'(a^*) > 0$, making the marginal cost of AI higher. Hence $a^* < a^{\text{no-debt}}$. $\square$

# C Proof of Proposition 2

**Part (i): Subjective risk is decreasing.** From (6) with $\mathbf{1}\{\text{crisis}_t\} = 0$ for all $t \in \mathcal{T}$: $\hat{\pi}_{t+1} = (1-\lambda)\hat{\pi}_t < \hat{\pi}_t$ for $\lambda \in (0,1)$. $\square$

**Part (ii): AI substitution intensity is increasing.** By part (i), $\hat{\pi}_t$ is strictly decreasing. By Proposition 1(ii), $\partial a^*/\partial \hat{\pi} < 0$. Hence $a_t^* = a^*(\hat{\pi}_t, q)$ is strictly increasing in $t$ along $\mathcal{T}$, holding $q$ fixed. $\square$

**Part (iii): Aggregate cognitive leverage is increasing.** The aggregate dynamics are:

$$B_{t+1} = (1+r_b)B_t + d(a_t^*), \qquad K_{t+1} = (1-\delta)K_t + \ell(1-a_t^*).$$

Compute:

$$\begin{aligned} \Omega_{t+1} &= \frac{(1+r_b)B_t + d(a_t^*)}{(1-\delta)K_t + \ell(1-a_t^*)} \\ &= \Omega_t \cdot \frac{(1+r_b)}{(1-\delta)} \cdot \frac{1 + d(a_t^*)/[(1+r_b)B_t]}{1 + \ell(1-a_t^*)/[(1-\delta)K_t]}. \end{aligned} \tag{27}$$

Since $r_b > \delta$ (cognitive debt compounds faster than cognitive capital depreciates naturally—an assumption we make explicit below) and $d(a_t^*)/B_t$ is positive, $\Omega_{t+1}/\Omega_t > 1$. Moreover, as $a_t^*$ increases, $d(a_t^*)$ rises and $\ell(1 - a_t^*)$ falls, further increasing $\Omega_{t+1}$. Hence $\Omega_t$ is strictly increasing. $\square$

**Assumption 5.** *$r_b > \delta$ and there exists $\bar{a} < 1$ such that for $a \geq \bar{a}$, $\ell(1-a)/K_t < r_b \cdot \Omega_t$ (the cognitive capital accumulation from practice does not offset leverage compounding at high substitution rates).*

**Part (iv): True crisis probability is increasing.** Since $\Pi' > 0$ and $\Omega_t$ is strictly increasing by part (iii), $\pi_t = \Pi(\Omega_t)$ is strictly increasing. $\square$

**Part (v): Minsky divergence.** From part (i): $\hat{\pi}_t = (1 - \lambda)^t \hat{\pi}_0 \to 0$ as $t \to \infty$. From part (iv): $\pi_t = \Pi(\Omega_t) \to \Pi(\infty) = 1$ as $t \to \infty$ (since $\Omega_t \to \infty$ when $r_b > \delta$ and $d(a^*) > 0$). Since $\hat{\pi}_t \to 0$ and $\pi_t \to 1$, there exists $T^* < \infty$ such that for all $t > T^*$, $\hat{\pi}_t < \pi_t$. $\square$

# D Proof of Proposition 3

Write $L_t = \Pi(\Omega_t) \cdot \Lambda(\Omega_t)$ where $\Lambda(\Omega) = \kappa\, \mathbb{E}_z[\max\{0, zB - \mathcal{V}(K)\}]$ and $B = \Omega K$ (by definition of $\Omega$).

**Part (i): $\partial L_t/\partial \Omega_t > 0$.** Both $\Pi(\Omega)$ and $\Lambda(\Omega)$ are positive and strictly increasing in $\Omega$ (higher leverage raises crisis probability and, conditional on crisis, raises the unsatisfied debt exposure). Their product is therefore strictly increasing. $\square$

**Part (ii): Convexity.** $L_t' = \Pi'\Lambda + \Pi\Lambda'$, and $L_t'' = \Pi''\Lambda + 2\Pi'\Lambda' + \Pi\Lambda''$.

We show $\Lambda'' > 0$. Consider $\Lambda(\Omega) = \kappa\,\mathbb{E}_z[z\Omega K - K^\alpha]_+$. For large enough $z$ (above the threshold $z^*(\Omega) = K^{\alpha-1}/\Omega$), the loss is in the interior. Differentiating:

$$\Lambda'(\Omega) = \kappa\,\mathbb{E}_z\left[zK \cdot \mathbf{1}\{z > z^*(\Omega)\}\right] > 0.$$

$$\Lambda''(\Omega) = \kappa\,\mathbb{E}_z\left[zK \cdot (-1) \cdot \frac{\partial z^*}{\partial \Omega} f_Z(z^*) + zK \cdot \mathbf{1}\{z > z^*\} \cdot 0\right] = \kappa\, K\, z^*(\Omega)\cdot(-\partial z^*/\partial\Omega)\cdot f_Z(z^*(\Omega)).$$

Since $z^*(\Omega) = K^{\alpha-1}/\Omega$, we have $\partial z^*/\partial\Omega = -K^{\alpha-1}/\Omega^2 < 0$, so $-\partial z^*/\partial\Omega > 0$, yielding $\Lambda''(\Omega) > 0$.

With $\Lambda'' > 0$, $\Lambda' > 0$, $\Pi' > 0$, and $\Pi \geq 0$: $L'' = \Pi''\Lambda + 2\Pi'\Lambda' + \Pi\Lambda'' > 2\Pi'\Lambda' > 0$ in any region where $-\Pi''\Lambda < 2\Pi'\Lambda'$. For the parametric form $\Pi(\Omega) = 1 - e^{-\lambda_\pi \Omega^\gamma}$ with $\gamma \geq 1$: $\Pi'' = \lambda_\pi e^{-\lambda_\pi\Omega^\gamma}[\lambda_\pi\gamma^2\Omega^{2\gamma-2} - \gamma(\gamma-1)\Omega^{\gamma-2}]$, which may be positive (when $\gamma > 1$), so that $L'' > 0$ globally. $\square$

**Part (iii): Model concentration.** Higher AI model concentration $M_t$ (HHI) raises the correlation of AI errors across agents. In a correlated error model, $\mathrm{Var}(z_t|M_t)$ increases in $M_t$, shifting the distribution of losses to the right. Since $\Lambda$ is convex in $z$ (the loss is linear for $z > z^*$ and zero otherwise), $\mathbb{E}[\Lambda(z)] \geq \Lambda(\mathbb{E}[z])$ by Jensen's inequality, and the inequality tightens as variance increases. Hence $\partial L_t/\partial M_t > 0$. $\square$

# E Proof of Proposition 4

After the crisis at $t = 0$, the agent in $t = 1$ chooses $a_1$ subject to the output constraint $y_{i1} \geq w$. With the constraint binding, we form the Lagrangian:

$$\mathcal{L}(a_1) = V_1(k_1', b_1') + \lambda_y\left[k_1 G(a_1; q) - w\right].$$

The constrained FOC for $a_1$:

$$\lambda_y \, k_1 \, G_a(a_1; q) = \beta \left[\mu_k \, \ell'(1 - a_1) + \mu_b \, d'(a_1)\right]. \tag{28}$$

Compare with the unconstrained pre-crisis FOC (12), which has $k_0 > k_1$ (cognitive capital has declined) and no $\lambda_y$ factor. The solution $a_1^*$ to (28) is higher than $a_0^-$ if and only if:

$$\lambda_y \, k_1 \, G_a(a_0^-; q) > \beta \left[\mu_k \, \ell'(1 - a_0^-) + \mu_b \, d'(a_0^-)\right],$$

which is condition (16). This follows because at $a_0^-$, the left-hand side of the constrained FOC (28) exceeds the right-hand side, implying the objective is still increasing in $a$, so the optimum $a_1^* > a_0^-$. □

With $a_1^* > a_0^-$: by the dynamics (3)–(4), $K_2 < K_1$ and $B_2 > B_1$, so $\Omega_2 > \Omega_1$. Applying this argument inductively, $\Omega_t$ is increasing for all $t \geq 1$, and $\pi_t = \Pi(\Omega_t)$ is increasing. Each successive crisis therefore occurs at a higher leverage level, producing weakly greater expected losses (by Proposition 3). □

# F Proof of Proposition 5

**Step 1: Derive the planner's FOC.** The planner maximises (18), treating $\pi_t = \Pi(\Omega_t)$ as endogenous. The FOC for $A_t$ includes the term $\partial/\partial A_t\big[\Pi(\Omega_t) \cdot L_t^{\text{cond}}\big]$ (where $L_t^{\text{cond}}$ is the expected conditional loss), which the decentralised agent omits. Expanding:

$$\Pi'(\Omega_t) \, \frac{\partial \Omega_t}{\partial A_t} \, L_t^{\text{cond}} > 0.$$

The planner also internalises $\partial K_{t+1}/\partial A_t = -\ell'(1 - A_t) < 0$ as a social cost (not just a private cost), and the benchmark-shifting externality $\partial \bar{y}_t / \partial a_{it}$.

**Step 2: Wedge between planner and decentralised FOCs.** In the decentralised equilibrium, the agent's FOC (12) is:

$$k\,\mathbb{E}\big[G_a^{\text{eff}}(a^*;\hat{\pi},q)\big] = \beta\left[\mu_k^D\,\ell'(1-a^*) + \mu_b^D\,d'(a^*)\right], \tag{FOC-D}$$

where $\mu_k^D, \mu_b^D$ are private shadow values that omit the externalities.

The planner's FOC is:

$$k\,\mathbb{E}\big[G_a^{\text{eff}}(A^P;\hat{\pi},q)\big] = \beta\left[\mu_k^P\,\ell'(1-A^P) + \mu_b^P\,d'(A^P)\right], \tag{FOC-P}$$

where $\mu_k^P > \mu_k^D$ (planner values cognitive capital more, accounting for the public good) and the left-hand side includes the systemic risk term.

The higher right-hand side in (FOC-P) implies $A^P < A^D$. $\square$

**Step 3: Pigouvian tax.** The optimal tax $\tau_t^*$ is set so that the decentralised agent, facing the after-tax problem, replicates the planner's FOC. The required tax on AI substitution intensity $a_{it}$ is the sum of the three marginal external costs evaluated at $(A_t^P, X_t^P)$:

1. Systemic risk: $\Pi'(\Omega_t)\,(\partial\Omega_t/\partial A_t)\,L_t/\Pi(\Omega_t)$ (marginal increase in expected loss per unit of $A_t$).

2. Public goods: $\beta\,\mu_k\,|\partial K_{t+1}/\partial A_t| = \beta\,\mu_k\,\ell'(1-A_t^P)$.

3. Arms-race: $\lambda_y\,\partial\bar{y}_t/\partial a_{it}$ (marginal increase in the competitive benchmark imposed on others).

Summing these gives (19).

**Step 4: The gap $\Delta_t$ is increasing in $M_t$, tranquil period length, and $d''$.** All three factors lower the private cost of AI use (relative to social cost) or raise the systemic externality:

- Higher $M_t$: increases $\mathrm{Var}(z|\text{crisis})$, raising $L_t$ and the first term in (19), widening the wedge.
- Longer tranquil period: $\Omega_t$ is higher (by Proposition 2), raising all three externality terms.
- Higher $d''$: makes the private debt cost steeper, which actually reduces individual $a^*$, but also raises the social risk of Ponzi dynamics, widening the systemic externality.

□

# G Infinite-Horizon Characterisation

In the infinite-horizon problem (8), the value function $V(k, b; \hat{\pi}, q)$ satisfies the Bellman equation. We verify that the comparative statics of Proposition 1 extend to this setting.

**Lemma 1.** *Under Assumptions 1–3, the value function $V$ is continuously differentiable with $V_k > 0$ and $V_b < 0$.*

*Proof.* Standard application of the Theorem of the Maximum and the Benveniste–Scheinkman envelope theorem, given that $G$ and $d$ are $C^2$ and the state space is compact (after imposing appropriate bounds on $k$ and $b$). □ □

The envelope conditions yield:

$$\mu_k^t \equiv V_k(k_t, b_t) = (1-\delta)\beta\, V_k(k_{t+1}, b_{t+1}) + \text{marginal product of } k_t \tag{29}$$

$$\mu_b^t \equiv -V_b(k_t, b_t) = (1+r_b)\beta\,(-V_b(k_{t+1}, b_{t+1})) + \kappa\,\hat{\pi}_t\,\bar{z} \tag{30}$$

These are the dynamic counterparts of $\mu_k$ and $\mu_b$ in the two-period model. The FOC (12) holds at each date with $\mu_k = \mu_k^t$ and $\mu_b = \mu_b^t$, confirming that the two-period characterisation applies period by period in the infinite-horizon setting. The comparative statics are therefore identical to those derived in Appendix B.

# H Proof of Proposition 6

**Part (i): $\Delta k_t$ is strictly decreasing.** The cognitive capital of each type evolves as:

$$k_{H,t+1} = (1-\delta)\, k_{H,t} + \ell(1 - a^*_{H,t}),$$
$$k_{L,t+1} = (1-\delta)\, k_{L,t} + \ell(1 - a^*_{L,t}).$$

Taking the difference:

$$\Delta k_{t+1} = (1-\delta)\,\Delta k_t + \ell(1 - a^*_{H,t}) - \ell(1 - a^*_{L,t}).$$

Since $a^*_{H,t} > a^*_{L,t}$ (from (20) while $\Delta k_t > 0$) and $\ell'(\cdot) > 0$, we have $1 - a^*_{H,t} < 1 - a^*_{L,t}$ and thus $\ell(1 - a^*_{H,t}) < \ell(1 - a^*_{L,t})$. Hence:

$$\Delta k_{t+1} < (1-\delta)\,\Delta k_t < \Delta k_t.$$

$\Delta k_t$ is strictly decreasing. $\square$

**Part (ii): $\Delta a_t^*$ converges to zero.** By the implicit function theorem applied to the FOC (12):

$$\frac{\partial a^*}{\partial k} = -\frac{\partial H/\partial k}{\partial H/\partial a} = -\frac{(1-\hat{\pi})G_a + \hat{\pi}\tilde{G}_a}{\partial H/\partial a} > 0.$$

As $\Delta k_t \to 0$, the types become identical, so $\Delta a_t^* \to 0$. □

**Part (iii): Reversal of fortune.** The gap dynamics satisfy $\Delta k_{t+1} = (1-\delta)\Delta k_t - [\ell(1-a_{L,t}^*) - \ell(1-a_{H,t}^*)]$.

By the mean value theorem: $\ell(1-a_{L,t}^*) - \ell(1-a_{H,t}^*) = \ell'(\xi_t)\,\Delta a_t^*$ for some $\xi_t \in (1-a_{H,t}^*, 1-a_{L,t}^*)$.

If condition (21) holds (the erosion differential exceeds the natural decay-corrected gap), then:

$$\Delta k_{t+1} = (1-\delta)\Delta k_t - \ell'(\xi_t)\Delta a_t^* < 0 \cdot \Delta k_t$$

when $\Delta k_t$ is close to zero from above. By continuity, $\Delta k_t$ must cross zero at some finite $T^{**}$. After crossing, the roles reverse: $k_{H,t} < k_{L,t}$, and—since $\partial a^*/\partial k > 0$—now $a_{H,t}^* < a_{L,t}^*$, making the gap $|\Delta k_t|$ persistent in the new direction. □

**Part (iv): Heterogeneity amplifies aggregate losses.** Expected aggregate loss is:

$$L_t^{\text{hetero}} = \tfrac{1}{2}\,\Pi(\Omega_{H,t})\,\Lambda(\Omega_{H,t}) + \tfrac{1}{2}\,\Pi(\Omega_{L,t})\,\Lambda(\Omega_{L,t}).$$

By Proposition 3, $L(\Omega) \equiv \Pi(\Omega)\Lambda(\Omega)$ is convex. Jensen's inequality applied to the two-point distribution of $\Omega_t$ gives:

$$L_t^{\text{hetero}} = \tfrac{1}{2}[L(\Omega_{H,t}) + L(\Omega_{L,t})] > L\left(\tfrac{\Omega_{H,t}+\Omega_{L,t}}{2}\right) = L_t^{\text{homo}},$$

where the last equality holds because the homogeneous economy has leverage $(\Omega_{H,t} + \Omega_{L,t})/2$ (since both types start with equal means and the distributions are mirror images around $\bar{k}$). $\square$

# I Shadow Value Derivation

We derive $\mu_k$ and $\mu_b$ used in the two-period model from the steady-state conditions of the infinite-horizon problem.

In the stationary equilibrium with constant $a, x, k, b$ and subjective probability $\hat{\pi}$:

$$\mu_k = \frac{(1-\hat{\pi})\,G(a;q) + \hat{\pi}\,\tilde{G}(a;q,\bar{z})}{1-\beta(1-\delta)}, \tag{31}$$

$$\mu_b = \frac{\hat{\pi}\,\kappa\,\bar{z}}{1-\beta(1+r_b)}. \tag{32}$$

Note that $\mu_b > 0$ requires $\beta(1+r_b) < 1$, which we assume. This condition says the debt cannot grow faster than the discount rate—otherwise cognitive debt would be "free" and Ponzi cognition would be trivially optimal.

Substituting (31)–(32) into (12) yields the explicit steady-state condition relating $a^*$ to the structural parameters $(\hat{\pi}, q, k, \beta, \delta, r_b, \kappa, \bar{z})$.